

Variance Theory: A Transition-Kernel Framework for Time, Causality, Complexity, Energy, Intelligence, and Open-Ended Persistence

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Abstract

This paper develops a revised formulation of **Variance Theory**: a formal framework in which time, causality, complexity, energy-constrained computation, and intelligence are derived from a single primitive: **state transition under physically constrained variance**. Earlier formulations treated the transition operator δ primarily as a deterministic state-transition function; the present paper corrects this by replacing δ with a **progressively randomizing transition kernel** whose deterministic, chaotic, epistemically stochastic, and truly random components are distinguished by conditional algorithmic incompressibility. Time is defined as a transition-cost quasi-metric induced by the support and expected hitting structure of the transition kernel. Causality is defined as the directed topology of possible transitions. Complexity is not identified with raw Kolmogorov complexity alone, because both periodic order and pure randomness are structurally shallow; instead, organized complexity is located in the region between determinism and randomness, corresponding to logical depth, effective complexity, statistical complexity, and the computational “edge of chaos.” Intelligence is formalized as the capacity of an adaptive policy to compress, predict, and exploit variance under energy, time, risk, and robustness constraints, thereby expanding a Pareto frontier rather than optimizing a single scalar objective.

A major correction concerns teleology. Variance Theory does not claim that the universe has a purpose. Instead, it proposes a persistence-selection principle: over long evolutionary horizons, structures that more effectively compress, predict, and utilize variance are statistically more likely to persist, reproduce, or recursively stabilize themselves. This is an ex post selection principle, not a cosmic intention. A second major correction concerns energy. Energy is not treated metaphorically; it is grounded in

physics as the generator, constraint, and thermodynamic cost of state transitions. In classical mechanics, Hamiltonians generate flows on phase space; in quantum mechanics, Hamiltonians generate unitary time evolution; by Noether's theorem, time-translation symmetry yields energy conservation; and by Landauer's principle, logically irreversible information processing has a lower thermodynamic cost. Thus, energy is the physically conserved capacity and cost structure through which variance is realized.

The paper proves three conditional results: a Persistence Selection Lemma, a No Final Compression Theorem, and a Pareto-Front Expansion Theorem for intelligence. It concludes that intelligence is not the attainment of a final state of complete knowledge but the ongoing expansion of feasible compression-prediction-control trade-offs within a world that contains irreducible residual variance at physical, computational, logical, and informational levels.

Keywords: Variance Theory; transition kernel; energy; Hamiltonian dynamics; Noether theorem; Landauer principle; Kolmogorov complexity; logical depth; edge of chaos; causal topology; Pareto frontier; intelligence; computational irreducibility; open-ended evolution.

1. Introduction

1.1 Motivation

Time, causality, energy, complexity, and intelligence are usually treated as belonging to separate domains: time and energy to physics, causality to metaphysics or causal inference, complexity to dynamical systems and information theory, and intelligence to cognitive science or artificial intelligence. Variance Theory proposes a more unified question:

What if these notions are different projections of the same primitive operation: physically constrained state transition?

The present paper revises and strengthens an earlier manuscript on Variance Theory, which introduced variance Υ and the transition function δ as a common basis for time, causality, intelligence, and the perpetuity of existence. The revision is substantial. The original deterministic formulation of δ is replaced by a transition-kernel formulation; the earlier language of cosmic purpose is removed; the notion of energy is physically grounded; and complexity is reformulated to avoid identifying pure randomness with organized complexity.

The central thesis is:

Variance is the physically realized transition of distinguishable states. Time is the metric of transition, causality is the topology of transition, complexity is the

organized accumulation of transition structure, energy is the physical generator and cost constraint of transition, and intelligence is the adaptive expansion of compression-prediction-control possibilities under those constraints.

This thesis should not be read as a claim that the universe “wants” intelligence. It is not teleological. The claim is instead selection-theoretic: structures that are better able to compress regularities, predict future transition structure, and exploit variance under physical constraints tend, over long horizons, to persist more effectively than structures that cannot.

1.2 Corrections to the Earlier Formulation

The present paper corrects four core issues.

First, δ must not be restricted to a deterministic function $\delta: \mathcal{S} \rightarrow \mathcal{S}$. Deterministic dynamics, chaotic dynamics, quantum measurement, environmental decoherence, hidden variables, and stochastic coarse-grained models cannot be handled rigorously by one deterministic signature. The correct object is a **transition kernel**:

$$P_{\delta}(ds' \mid s, a),$$

which assigns a probability measure over next states s' conditional on a current state s and an optional action a . Deterministic dynamics appear as the degenerate case

$$P_{\delta}(ds' \mid s, a) = \mathbf{1}_{F(s,a)}(ds').$$

Second, randomness is not treated as a binary switch. We introduce **progressive true randomness**: as a system improves its model class, some apparent randomness may be compressed away as hidden structure, but a residual may remain. The limit of this residual under expanding admissible model classes distinguishes true randomness from ignorance.

Third, the framework must avoid strong teleology. It does not claim that the universe has a purpose. It claims that long-lived structures are selected by their capacity to preserve organization under variance. This is analogous in form to Darwinian and replicator-selection reasoning but generalized to compression, prediction, utilization, and persistence.

Fourth, energy must be connected to physics. Energy is not merely “motivation” or “force.” In Hamiltonian mechanics, it is encoded in the Hamiltonian that generates phase-space flow; in quantum mechanics, the Hamiltonian generates time evolution through the Schrödinger equation; by Noether’s theorem, time-translation symmetry yields energy conservation; by Landauer’s principle, irreversible information processing requires thermodynamic dissipation. Noether’s theorem connects continuous symmetries of the action with conservation laws, and time-translation symmetry is the standard route to energy conservation. ([arXiv](#)) Landauer’s 1961 result connects logical irreversibility with physical heat generation, and later experiments verified the Landauer bound in one-bit erasure systems. ([IEEE Xplore](#))

2. Formal Preliminaries

2.1 State Space, Actions, Histories, and Transition Kernels

Let

$$\mathcal{S}$$

be a measurable state space. A state $s \in \mathcal{S}$ may represent a microscopic physical state, a macroscopic coarse-grained state, a computational configuration, an organism-environment state, or an agent-in-world state. Let

$$\mathcal{A}$$

be an action space. For closed physical systems, $\mathcal{A}$ may be a singleton or absent. For adaptive systems, actions represent interventions, control choices, learning updates, attention allocation, or self-modifications.

A **history** up to time t is

$$h_t = (s_0, a_0, s_1, a_1, \dots, s_t).$$

The basic object of Variance Theory is not a deterministic function but a transition kernel:

$$P_\delta(\cdot | s_t, a_t, h_t) \in \Delta(\mathcal{S}),$$

where $\Delta(\mathcal{S})$ denotes the set of probability measures over $\mathcal{S}$.

For Markovian systems,

$$P_\delta(\cdot | s_t, a_t, h_t) = P_\delta(\cdot | s_t, a_t).$$

For non-Markovian systems, the kernel may depend on the whole history h_t .

A deterministic transition function is recovered when there exists a measurable map $F: \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$ such that

$$P_\delta(B | s, a) = \begin{cases} 1, & F(s, a) \in B, \\ 0, & F(s, a) \notin B. \end{cases}$$

Thus deterministic δ is a special case of the broader transition-kernel formalism.

2.2 Variance

Definition 1. Variance.

A system exhibits variance if there exist states $s, s' \in \mathcal{S}$ and an action or admissible condition $a \in \mathcal{A}$ such that

$$P_\delta(s' \neq s \mid s, a) > 0.$$

In continuous spaces, this becomes

$$P_\delta(\mathcal{S} \setminus s \mid s, a) > 0.$$

A perfectly static system has

$$P_\delta(s \mid s, a) = 1$$

for all s, a . Such a system has no physical time readout, no causal graph beyond identity, and no nontrivial dynamics.

2.3 Coarse-Graining and Observer-Relative State Spaces

Let $\mathcal{X}$ be a microscopic state space and let

$$\Pi: \mathcal{X} \rightarrow \mathcal{S}$$

be a coarse-graining map. A macrostate $s \in \mathcal{S}$ corresponds to the set

$$\Gamma(s) = \{x \in \mathcal{X} : \Pi(x) = s\}.$$

Even if microscopic dynamics are reversible, the induced macroscopic transition kernel

$$P_{\delta, \Pi}(s' \mid s)$$

may be many-to-one because many microstates in $\Gamma(s)$ may flow into a macroregion $\Gamma(s')$. Decoherence provides a central physical example: environmental interaction suppresses interference between pointer states and yields effective classicality for open subsystems, even when global quantum dynamics are unitary. ([arXiv](#))

This distinction matters because Variance Theory is operationally formulated over physically accessible state spaces, not over a hypothetical omniscient description of the universal wavefunction.

3. Progressive True Randomness

3.1 Why Randomness Must Be Progressive

A sequence that appears random to a primitive observer may become compressible to a more powerful observer. Conversely, a sequence that remains incompressible even under unbounded improvement of admissible physical models is a candidate for true randomness. Therefore, randomness should not be defined relative to one fixed model. It should be defined through a progressive sequence of model classes.

Let

$$\mathcal{M}_0 \subseteq \mathcal{M}_1 \subseteq \dots \subseteq \mathcal{M}_\infty$$

be an increasing hierarchy of admissible models. Here $\mathcal{M}_0$ may contain simple deterministic rules; $\mathcal{M}_1$ may contain chaotic models; $\mathcal{M}_2$ may contain stochastic coarse-grained models; and $\mathcal{M}_\infty$ denotes the limiting class of all physically admissible computable models.

For a next-state observation s_{t+1} , define the residual algorithmic surprise relative to model class $\mathcal{M}_n$:

$$R_t^{(n)} = \inf_{M \in \mathcal{M}_n} K(s_{t+1} \mid h_t, M),$$

where $K(\cdot \mid \cdot)$ is conditional Kolmogorov complexity. Kolmogorov's algorithmic approach defines information by shortest effective description length, and the incompressibility of typical strings follows from this framework. ([Knownen](#))

Because the model classes are nested,

$$R_t^{(0)} \geq R_t^{(1)} \geq R_t^{(2)} \geq \dots$$

The limiting residual is

$$R_t^{(\infty)} = \lim_{n \rightarrow \infty} R_t^{(n)}.$$

Definition 2. Progressive true-randomness rate.

$$\rho_{\text{TR}} = \liminf_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \frac{R_t^{(\infty)}}{\ell(s_{t+1})},$$

where $\ell(s_{t+1})$ is the description length of the observed next state.

If $\rho_{\text{TR}} > 0$, the system contains a positive asymptotic rate of incompressible residual variance. If $\rho_{\text{TR}} = 0$ but prediction still requires stepwise execution, the system may be deterministic but computationally irreducible. Wolfram's work on undecidability and intractability in theoretical physics argues that many physical questions may require irreducible computation rather than closed-form shortcuts. ([Wolfram Content](#))

3.2 Four Kinds of Apparent Randomness

Variance Theory distinguishes four levels.

Level R0: Deterministic regularity.

$$P_{\delta}(s' | s, a) = \mathbf{1}_{F(s,a)}(s').$$

The next state is fully determined by a computable map.

Level R1: Deterministic chaos.

The system is deterministic but sensitive to initial conditions. Apparent unpredictability arises from finite measurement precision, not necessarily from true randomness.

Level R2: Effective stochasticity.

A coarse-grained or open-system description is stochastic because the model omits hidden or environmental degrees of freedom. Decoherence and open-system thermodynamics often yield such effective kernels. ([APS Links](#))

Level R3: True residual randomness.

Even after all physically admissible compression is exhausted, a positive residual remains:

$$\rho_{\text{TR}} > 0.$$

Objective-collapse interpretations of quantum mechanics would make this physically fundamental; unitary interpretations may preserve it only as branch-relative or epistemic incompressibility for embedded observers. Collapse models are experimentally motivated stochastic modifications of quantum theory, while decoherence explains effective classicality without necessarily postulating fundamental collapse. ([APS Links](#))

3.3 The Progressively Randomizing Kernel

The revised transition kernel can be decomposed as

$$P_{\delta} = (1 - \alpha - \beta - \gamma)P_{\text{det}} + \alpha P_{\text{chaos}} + \beta P_{\text{eff}} + \gamma P_{\text{TR}},$$

where

$$\alpha, \beta, \gamma \geq 0, \alpha + \beta + \gamma \leq 1.$$

Here:

- P_{det} is the deterministic law-like component;
- P_{chaos} represents deterministic sensitivity under finite precision;
- P_{eff} represents stochasticity from coarse-graining and environmental omission;
- P_{TR} represents irreducible true-random residuals.

The coefficients are not universal constants. They are scale-, observer-, and model-dependent. As an intelligent system improves its models, some mass may move from P_{eff} or P_{chaos} into P_{det} . What cannot be moved in the limit contributes to P_{TR} .

Thus true randomness is not assumed dogmatically. It is defined as the limiting residue of all admissible compression.

4. Energy as the Physical Driver, Generator, Constraint, and Cost of Variance

4.1 Energy Must Not Be Metaphorical

The phrase “energy is the driver of variance” is acceptable only if made physically precise. In the revised framework, energy has four roles:

1. **Generator:** it enters the laws that generate state transitions;
2. **Constraint:** it restricts accessible state space;
3. **Currency:** it measures the capacity to perform work;
4. **Cost:** it bounds irreversible information processing.

These are not metaphors. They correspond to standard structures in mechanics, quantum theory, and thermodynamics.

4.2 Classical Hamiltonian Dynamics

Let $(\mathcal{P}, \omega)$ be a symplectic phase space and let

$$H: \mathcal{P} \rightarrow \mathbb{R}$$

be a Hamiltonian. The Hamiltonian vector field X_H is defined by

$$\iota_{X_H} \omega = dH.$$

The classical state evolves according to

$$\dot{z} = X_H(z).$$

The flow generated by H is

$$\Phi_H^t: \mathcal{P} \rightarrow \mathcal{P}.$$

Therefore the deterministic transition map at time step Δt is

$$\delta_{\Delta t}^H(z) = \Phi_H^{\Delta t}(z).$$

If $X_H = 0$ on a region, states in that region do not undergo nontrivial Hamiltonian transition. If $X_H \neq 0$, H generates variance through phase-space flow. This is the strict sense in which energy, as encoded by the Hamiltonian, drives physically realized transition.

4.3 Quantum Time Evolution

In quantum mechanics, a state vector evolves according to the Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle.$$

For time-independent $\hat{H}$,

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle, U(t) = e^{-i\hat{H}t/\hbar}.$$

Thus the Hamiltonian operator $\hat{H}$ generates unitary time evolution. Standard quantum lecture notes and textbooks formulate time evolution in exactly this Hamiltonian-generated way. ([Ethan Neil](#))

The corresponding transition kernel is deterministic at the level of the closed quantum state:

$$P_\delta(d\psi' | \psi) = \mathbf{1}_{U(t)\psi}(d\psi').$$

However, for measurements, open systems, or branch-relative observers, the effective transition over observed outcomes is generally stochastic.

4.4 Energy Conservation and Noether's Theorem

Noether's theorem states that continuous symmetries of the action correspond to conserved quantities. In particular, invariance under time translation yields conservation of energy in standard Lagrangian systems. ([arXiv](#))

Let the action be

$$S[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t), dt.$$

If

$$\frac{\partial L}{\partial t} = 0,$$

then the energy function

$$E = \sum_i \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - L$$

is conserved along solutions of the Euler-Lagrange equations.

This gives the corrected interpretation:

Energy is not “used up” as variance occurs. In closed time-translation-invariant systems, energy is conserved while generating or constraining the admissible transition flow. Thus energy is both the generator of transition and a conserved constraint on transition.

4.5 Thermodynamic and Informational Cost

For computation and intelligence, energy is also a cost. Landauer’s principle states that logically irreversible operations, such as bit erasure, have a minimum thermodynamic cost of order

$$k_B T \ln 2$$

per erased bit under ideal conditions. Landauer originally connected logical irreversibility with heat generation in computing machines, and later experiments directly verified the Landauer bound in physical one-bit memories. ([IEEE Xplore](#))

Let J_{erase} denote the number of bits irreversibly erased along a computational transition path γ . Then the thermodynamic cost satisfies

$$Q_\gamma \geq k_B T \ln 2 \cdot J_{\text{erase}}.$$

This matters for intelligence. Compression, prediction, memory revision, search, and action selection are not abstract operations floating outside physics. They are physical computations with thermodynamic costs.

4.6 Stochastic Thermodynamics

For mesoscopic systems, stochastic thermodynamics extends work, heat, and entropy production to individual fluctuating trajectories. It applies to systems coupled to thermal

reservoirs and often imposes local detailed-balance constraints on transition rates. ([arXiv](#))

For a Markov jump process with rates W_{ij} , local detailed balance often has the form

$$\ln \frac{W_{ij}}{W_{ji}} = \beta q_{ij},$$

where q_{ij} is heat exchanged with the bath and $\beta = (k_B T)^{-1}$.

This makes the energy-variance link exact in stochastic settings: transition probabilities are constrained by thermodynamic exchange.

4.7 Definition of Energetic Variance Cost

For a trajectory

$$\gamma = (s_0, a_0, s_1, \dots, s_T),$$

define the energetic variance cost:

$$E_\delta(\gamma) = W_{\text{in}}(\gamma) + Q_{\text{diss}}(\gamma) + E_{\text{maint}}(\gamma),$$

where:

- W_{in} is external work input;
- Q_{diss} is dissipated heat;
- E_{maint} is the energetic cost of preserving structure away from equilibrium.

For computational systems, include Landauer cost:

$$E_\delta(\gamma) \geq k_B T \ln 2 \cdot J_{\text{erase}}(\gamma).$$

Thus, energy in Variance Theory is rigorously defined as the physical generator, conservation constraint, and thermodynamic price of transition.

5. Time as the Quasi-Metric of Variance

5.1 Support Reachability

Given a transition kernel P_δ , define the support edge relation:

$$s \rightarrow_\delta s' \Leftrightarrow \exists a \in \mathcal{A} \text{ such that } P_\delta(U_{s'} \mid s, a) > 0$$

for every sufficiently small neighborhood $U_{s'}$ of s' .

The directed support graph is

$$\mathcal{G}_\delta = (\mathcal{S}, E_\delta),$$

where

$$E_\delta = (s, s') : s \rightarrow_\delta s'.$$

5.2 Transition Distance

For deterministic systems, the earlier distance definition remains valid:

$$d_\delta(s_i, s_j) = \min n \in \mathbb{N}_0 : \delta^n(s_i) = s_j.$$

For stochastic systems, exact equality is usually too strong. We define the support distance:

$$d_{\text{sup}}(s_i, s_j) = \min n \in \mathbb{N}_0 : \exists \pi, ; P_\delta^\pi(s_n \in U_{s_j} \mid s_0 = s_i) > 0.$$

We also define the expected hitting-time distance:

$$d_{\text{hit}}(s_i, s_j) = \inf_\pi \mathbb{E}_\pi[\tau_{s_j} \mid s_0 = s_i],$$

where

$$\tau_{s_j} = \inf t : s_t \in U_{s_j}.$$

The time readout of variance is therefore not a primitive background container. It is the cost or length of transition:

$$T_\delta(s_i \rightarrow s_j) = d_\delta(s_i, s_j),$$

with the relevant version depending on whether the system is deterministic, stochastic, microscopic, macroscopic, exact, or coarse-grained.

5.3 Quasi-Metric Properties

A quasi-metric satisfies:

$$\begin{aligned} d(s, s) &= 0, \\ d(s_i, s_k) &\leq d(s_i, s_j) + d(s_j, s_k), \end{aligned}$$

but does not require symmetry:

$$d(s_i, s_j) \neq d(s_j, s_i).$$

In transition systems, asymmetry is natural. If many prior states converge into one macrostate through coarse-graining or irreversible dissipation, the forward transition may be easy while the reverse transition is not physically feasible without additional information and work.

Thus the arrow of time is not inserted by hand. It emerges from directed transition structure, coarse-graining, thermodynamic irreversibility, and entropy-producing dynamics.

5.4 Microscopic Reversibility and Macroscopic Irreversibility

Variance Theory does not require fundamental microscopic irreversibility. It requires only operationally relevant irreversibility at the level of accessible states.

Closed quantum evolution may be unitary and reversible. But an embedded observer generally has access only to a subsystem, a measurement record, or a coarse-grained macrostate. Environmental decoherence explains why classical pointer states become effectively stable and why interference becomes inaccessible at macroscopic scales. ([arXiv](#))

Thus the revised claim is:

Time's arrow is a property of effective transition structure for physically embedded observers, and may or may not be fundamental depending on the interpretation and dynamics of quantum theory.

This avoids the stronger and unnecessary claim that all fundamental δ must be non-injective.

6. Causality as the Topology of Variance

6.1 Causal Reachability

Define the causal preorder:

$$s_i \leq_{\delta} s_j \Leftrightarrow d_{\text{sup}}(s_i, s_j) < \infty.$$

This relation is reflexive and transitive. It may fail to be antisymmetric if cycles exist. If cycles are quotient out by the equivalence relation

$$s_i \sim s_j \Leftrightarrow s_i \preceq_{\delta} s_j \text{ and } s_j \preceq_{\delta} s_i,$$

then the quotient space $\mathcal{S}/\sim$ has a partial-order structure when no higher-order cycles remain.

This is the sense in which causality is the topology of variance: causality is the directed reachability structure generated by possible transitions.

6.2 Relation to Causal Set Theory

Causal Set Theory models spacetime at the deepest level as a locally finite partially ordered set, where the partial order corresponds to causal precedence and local finiteness encodes discreteness. Bombelli, Lee, Meyer, and Sorkin proposed that spacetime at small scales may be a causal set; Surya's review summarizes causal sets as locally finite posets whose order represents proto-causality. ([APS Links](#))

Variance Theory is compatible with this idea but differs in ontological emphasis.

Causal Set Theory begins with order:

$$(\mathcal{C}, \preceq).$$

Variance Theory begins with a transition kernel:

$$P_{\delta}(ds' \mid s, a),$$

and derives order through reachability:

$$s_i \preceq_{\delta} s_j \Leftrightarrow d_{\text{sup}}(s_i, s_j) < \infty.$$

Thus Variance Theory is a generative strengthening of causal topology. It asks not only which events are ordered, but what transition structure generates that order.

7. Complexity: Between Determinism and Randomness

7.1 Why Kolmogorov Complexity Is Not Enough

Kolmogorov complexity $K(x)$ measures the length of the shortest program that outputs x . It is foundational for formal incompressibility. ([Knownen](#))

However, raw $K(x)$ does not by itself capture organized complexity.

Consider three sequences:

1. Periodic sequence:

0101010101 ...

It has low K and low organized complexity.

2. Unknown encrypted or truly random sequence:

$x \in 0,1^n, K(x) \approx n$.

It has high K but may be structurally useless to an observer without the key.

3. Biological, linguistic, technological, or cognitive structure.

It is neither trivially periodic nor maximally random. It contains regularities, hierarchy, constraints, memory, adaptive feedback, and nontrivial causal history.

Therefore, organized complexity is not maximal randomness. Bennett's logical depth was introduced precisely to distinguish shallow randomness from objects with nontrivial causal histories. ([UC Davis Computer Science](#)) Gell-Mann and Lloyd's effective complexity similarly targets the compressed description length of regularities rather than the random component itself. ([CaltechAUTHORS](#))

7.2 Edge of Chaos

Complex systems theory often focuses on the region between rigid order and unstructured randomness. Langton's work on computation at the edge of chaos argued that systems supporting emergent computation must allow transmission, storage, and modification of information, and that such capacities arise near phase-transition regimes rather than in fully ordered or fully disordered regimes. ([ScienceDirect](#))

Variance Theory therefore defines organized complexity as a balance between:

- compressible regularity;
- residual uncertainty;
- causal depth;
- manipulability;
- model-improvability.

A frozen crystal, a simple loop, and a periodic automaton are too determined. A one-time-pad encrypted sequence or truly random string is too unstructured relative to the observer. Intelligence emerges in the region where uncertainty remains but regularities can still be discovered and exploited.

7.3 Observer-Relative Complexity

Let M be an observer's model. Define observer-relative algorithmic complexity:

$$K_M(x) = \min |p| : U(p, M) = x.$$

An encrypted sequence may have low complexity relative to an observer with the key and high complexity relative to an observer without the key. Therefore, complexity in adaptive systems is partly observer-relative.

Define the compressible regularity of x relative to M :

$$\text{Reg}_M(x) = \ell(x) - K_M(x).$$

Define the residual unpredictability:

$$\text{Res}_M(x) = K_M(x | M).$$

A system is in the intelligence-relevant complexity zone when both terms are nonzero:

$$\text{Reg}_M(x) > 0, \text{Res}_M(x) > 0.$$

The first condition means there is structure to learn. The second means there remains uncertainty to explore.

7.4 Organized Variance Complexity

Let $\mathcal{G}_\delta[0, T]$ be the causal-transition graph observed over a horizon T . Define organized variance complexity:

$$\Omega_T = \text{EC}(\mathcal{G}_\delta[0, T]) \cdot \log(1 + \text{LD}(\mathcal{G}_\delta[0, T])),$$

where:

- EC is effective complexity: compressed regularity;
- LD is logical depth: minimal plausible generative computation time.

This definition intentionally penalizes both trivial order and shallow randomness. A periodic loop has low effective complexity and low depth. A random sequence has high raw Kolmogorov complexity but low recoverable regularity. Organized complex systems have nontrivial regularities generated through nontrivial causal histories.

7.5 Statistical Complexity

Crutchfield and Young introduced statistical complexity as a measure distinct from entropy and dimension, intended to capture the information-processing structure of nonlinear dynamical systems. ([APS Links](#)) Variance Theory is aligned with that

distinction: entropy rate measures unpredictability, but intelligence-relevant complexity concerns the internal causal architecture that supports prediction and control.

8. Intelligence as Compression, Prediction, Utilization, and Pareto-Front Expansion

8.1 Why Intelligence Is Not Scalar Optimization

A simple scalar definition such as

$$\eta = \frac{\text{problem complexity solved}}{\text{energy} \times \text{time}}$$

is useful but incomplete. Real intelligence involves trade-offs:

- accuracy versus energy;
- speed versus robustness;
- exploitation versus exploration;
- compression versus sensitivity to novelty;
- short-term reward versus long-term persistence;
- certainty versus adaptability.

Thus intelligence should be formulated as multi-objective optimization over a Pareto frontier. Multi-objective optimization studies cases where several objectives must be optimized simultaneously and where no single solution dominates all others across every criterion. ([College of Engineering](#))

8.2 Policy and Internal Model

Let an intelligent system have:

$$\pi(a_t \mid h_t, M_t),$$

where π is a policy and M_t is an internal model of the transition kernel.

The model M_t is used to produce predictions:

$$\hat{P}_t(s_{t+1} \mid h_t, a_t).$$

The system updates M_t through learning:

$$M_{t+1} = \mathcal{U}(M_t, h_t, a_t, s_{t+1}).$$

8.3 Compression Objective

Define cumulative compression gain:

$$C_T(\pi) = \sum_{t=0}^{T-1} [K(s_{t+1} | h_t, M_t^-) - K(s_{t+1} | h_t, M_t^+)],$$

where M_t^- is the model before learning and M_t^+ is the model after learning.

This measures how much the agent reduces description length by discovering structure.

8.4 Prediction Objective

Define predictive loss:

$$L_T(\pi) = \sum_{t=0}^{T-1} -\log \hat{P}_t(s_{t+1} | h_t, a_t).$$

Prediction quality is

$$P_T(\pi) = -L_T(\pi).$$

This connects directly to log-loss, Bayesian prediction, and algorithmic sequence prediction. Legg and Hutter define universal intelligence through expected reward across environments weighted by algorithmic simplicity, while Hutter's AIXI formalism connects sequential decision-making with algorithmic probability. ([arXiv](#))

8.5 Utilization Objective

Compression and prediction are not enough. Intelligence also uses variance.

Let $u(s, a)$ be task utility, survival utility, reproductive utility, or structural-stability utility. Define

$$U_T(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} \gamma^t u(s_t, a_t) \right].$$

This captures the exploitation of discovered regularities.

8.6 Energy, Time, Risk, and Robustness

Define energetic cost:

$$E_T(\pi) = \mathbb{E}_\pi[E_\delta(\gamma_T)].$$

Define transition-time cost:

$$T_T(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} c_T(s_t, a_t, s_{t+1}) \right].$$

Define risk:

$$R_T(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} r(s_t, a_t) \right].$$

Define robustness:

$$B_T(\pi) = -\text{Var}_\pi[U_T]$$

or, more generally, worst-case performance over perturbations:

$$B_T(\pi) = \inf_{P' \in \mathcal{B}(P_\delta)} U_T(\pi; P').$$

8.7 The Intelligence Vector

Define the intelligence-performance vector:

$$\mathbf{F}_T(\pi) = (C_T(\pi), P_T(\pi), U_T(\pi), B_T(\pi), -E_T(\pi), -T_T(\pi), -R_T(\pi)).$$

A policy π_1 Pareto-dominates π_2 if

$$F_i(\pi_1) \geq F_i(\pi_2) \forall i,$$

and

$$F_j(\pi_1) > F_j(\pi_2) \text{ for at least one } j.$$

The Pareto frontier is

$$\mathcal{P}_T = \{\pi : \nexists \pi' \text{ such that } \pi' \succ \pi\}.$$

8.8 Definition of Intelligence

Definition 3. Intelligence.

Intelligence is the capacity of a physical or computational system to expand the attainable Pareto frontier of compression, prediction, utilization, robustness, energy efficiency, temporal efficiency, and risk management over progressively uncertain transition structures.

Formally, let

$$\text{HV}(\mathcal{P}_T)$$

denote the hypervolume dominated by the Pareto frontier relative to a reference point. Then intelligence over horizon T is measured by

$$\mathcal{I}_T = \text{HV}(\mathcal{P}_T) / \text{HV}(\mathcal{P}_0).$$

This definition captures the user-level intuition:

Intelligence explores regularity inside uncertainty, compresses what can be compressed, predicts what can be predicted, exploits what can be used, and continuously expands the Pareto frontier of feasible trade-offs.

8.9 Relation to Existing Intelligence Theories

This formulation is compatible with but broader than several existing theories.

Legg-Hutter intelligence concerns expected reward across computable environments weighted by simplicity. ([arXiv](#)) Chollet’s measure of intelligence emphasizes skill-acquisition efficiency and generalization under limited priors and experience. ([arXiv](#)) Friston’s Free Energy Principle treats perception, action, and learning as forms of free-energy or surprise minimization. ([Nature](#))

Variance Theory generalizes these approaches by making surprise minimization only one objective among several. An intelligent system should not minimize surprise to zero, because doing so may eliminate exploration, novelty, and adaptation. Instead, intelligence manages the trade-off between compression and residual variance.

9. Anti-Teleology: The Persistence Selection Principle

9.1 Removing Strong Purpose

The revised framework rejects the claim:

The universe exists in order to increase problem-solving efficiency.
That claim is too strongly teleological and not required.

The corrected claim is:

Over long evolutionary horizons, structures that more effectively compress, predict, and utilize variance under energetic and temporal constraints are more likely to persist, reproduce, or recursively stabilize themselves.
This is a selection principle, not an intention principle.

9.2 Persistence Functional

Let X_i be a class of structures: molecules, organisms, minds, institutions, technologies, models, or civilizations. Define its persistence functional:

$$\Lambda(X_i) = \alpha C_i + \beta P_i + \gamma U_i + \eta B_i - \lambda_E E_i \lambda_T T_i \lambda_R R_i,$$

where:

- C_i is compression capacity;
- P_i is predictive capacity;
- U_i is utilization capacity;
- B_i is robustness;
- E_i is energetic cost;
- T_i is temporal cost;
- R_i is risk.

This is not a universal moral value or cosmic objective. It is an environment-relative statistical persistence score.

9.3 Persistence Selection Lemma

Suppose two structure classes X_i and X_j have frequencies $x_i(t)$ and $x_j(t)$ satisfying the replicator form

$$\begin{aligned}\dot{x}_i &= x_i(\Lambda_i - \bar{\Lambda}), \\ \dot{x}_j &= x_j(\Lambda_j - \bar{\Lambda}).\end{aligned}$$

Then

$$\frac{d}{dt} \ln \frac{x_i}{x_j} = \Lambda_i - \Lambda_j.$$

If

$$\Lambda_i - \Lambda_j \geq \epsilon > 0$$

over an interval $[0, T]$, then

$$\frac{x_i(T)}{x_j(T)} \geq \frac{x_i(0)}{x_j(0)} e^{\epsilon T}.$$

Proof.

$$\frac{d}{dt} \ln \frac{x_i}{x_j} = \frac{\dot{x}_i}{x_i} - \frac{\dot{x}_j}{x_j} = (\Lambda_i - \bar{\Lambda}) - (\Lambda_j - \bar{\Lambda}) = \Lambda_i - \Lambda_j.$$

Integrating over $[0, T]$ gives

$$\ln \frac{x_i(T)/x_j(T)}{x_i(0)/x_j(0)} = \int_0^T (\Lambda_i - \Lambda_j) dt \geq \epsilon T.$$

Exponentiating gives the result.

This lemma formalizes the corrected non-teleological claim. There is no cosmic purpose. There is only differential persistence under variance.

10. The Four Non-Closure Principles

Variance Theory identifies four independent reasons why complete closure is impossible or unstable.

10.1 Physical Layer: Progressive True Randomness

If

$$\rho_{\text{TR}} > 0,$$

then physical transition sequences contain irreducible residuals that cannot be compressed away by improved modeling. Whether this is fundamental or observer-relative depends on quantum interpretation. Collapse models would make randomness fundamental; unitary interpretations may make it branch-relative or epistemic for embedded observers. ([APS Links](#))

10.2 Computational Layer: Computational Irreducibility

Even when $\rho_{\text{TR}} = 0$, a deterministic process may be computationally irreducible. The future cannot be obtained by a shortcut substantially cheaper than executing the process. Wolfram’s work on physical undecidability and computational intractability provides a formal basis for this kind of irreducibility. ([Wolfram Content](#))

10.3 Logical Layer: Gödelian Incompleteness

For sufficiently expressive, consistent, effectively axiomatized formal systems, there exist true statements not provable within the system. Gödel’s incompleteness theorem blocks total self-closure for formal systems containing arithmetic. ([PhilArchive](#)) Chaitin later connected incompleteness with algorithmic information: formal systems of bounded complexity cannot prove arbitrarily high algorithmic randomness. ([ACM Digital Library](#))

For self-modeling intelligent systems, this means that a finite internal model cannot fully close over its own transition structure without generating new meta-level description demands.

10.4 Information Layer: Kolmogorov Incompressibility

Most strings of length n are incompressible by counting argument: there are fewer short programs than long strings. Kolmogorov complexity formalizes this residual. ([Known](#))

The informational layer is the most general: even where physical randomness is disputed, and even where deterministic rules exist, finite descriptions cannot compress all possible variance to zero.

11. No Final Compression Theorem

11.1 Statement

Let $\Sigma_T = (s_0, s_1, \dots, s_T)$ be a variance sequence generated by P_δ . Let $\mathcal{M}_T$ be the model class available to an intelligent system by time T .

Define residual complexity:

$$\mathcal{R}_T = K(\Sigma_T \mid \mathcal{M}_T).$$

Theorem 1. No Final Compression Theorem.

If either:

1. the progressive true-randomness rate satisfies $\rho_{\text{TR}} > 0$; or

2. the transition process is computationally irreducible;
then no finite model M can reduce all future variance to zero residual complexity at finite cost.

11.2 Proof

Case 1: Suppose $\rho_{\text{TR}} > 0$. By definition,

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \frac{R_t^{(\infty)}}{\ell(s_{t+1})} > 0.$$

Therefore, even under the limiting admissible model class, residual conditional description length accumulates at positive asymptotic rate. If a finite model M compressed all future variance to zero residual, then for sufficiently large T ,

$$K(\Sigma_T \mid M) = o(T),$$

contradicting the positive residual rate.

Case 2: Suppose $\rho_{\text{TR}} = 0$ but the process is computationally irreducible. Then the future state s_T may be determined by a finite rule, but no shortcut computes it substantially faster than executing the process. A finite model may describe the rule, but it cannot collapse the computational cost of future variance to zero. Thus all future variance cannot be pre-resolved at finite cost.

Therefore, under either condition, complete closure fails.

12. Pareto-Front Expansion Theorem

12.1 Statement

Let $\mathcal{P}_T$ be the Pareto frontier of policies available to an intelligent system at time T . Suppose learning expands the admissible policy-model set:

$$\Pi_T \subseteq \Pi_{T+1}.$$

Theorem 2. Pareto-Front Expansion Theorem.

If there exists a newly available policy $\pi' \in \Pi_{T+1}$ that Pareto-dominates at least one policy $\pi \in \mathcal{P}_T$, then the attainable frontier strictly expands.

12.2 Proof

Because $\Pi_T \subseteq \Pi_{T+1}$, every old feasible policy remains feasible. If π' dominates some $\pi \in \mathcal{P}_T$, then π is no longer nondominated in the enlarged feasible set. Since π' is feasible and improves at least one objective without worsening others, the dominated hypervolume relative to any fixed reference point strictly increases. Therefore,

$$\text{HV}(\mathcal{P}_{T+1}) > \text{HV}(\mathcal{P}_T).$$

Thus intelligence increases by frontier expansion.

This theorem captures the corrected essence of intelligence: not final optimization, but continuing expansion of feasible trade-offs.

13. The Role of the Edge of Chaos in Intelligence

13.1 Static Order

If a system is fully static,

$$P_\delta(s \mid s) = 1,$$

then prediction is trivial and no adaptive intelligence is required.

If a system is periodic,

$$\delta^k(s) = s,$$

then prediction remains highly compressible. Such systems can be modeled with short descriptions and do not require deep exploration.

13.2 Pure Randomness

If a sequence is maximally incompressible,

$$K(x) \approx \ell(x),$$

then there is little exploitable regularity. Prediction cannot improve beyond distributional constraints.

13.3 Edge-of-Chaos Region

Intelligence becomes meaningful where:

$$0 < \text{Reg}_M(x) < \ell(x),$$

and

$$0 < \text{Res}_M(x) < \ell(x).$$

There is enough order to compress, enough uncertainty to explore, enough stability to exploit, and enough novelty to require adaptation.

This aligns with Langton's edge-of-chaos thesis: computation requires a substrate that can transmit, store, and modify information, which tends to occur near transition regimes rather than at extremes of rigid order or maximal disorder. ([ScienceDirect](#))

Thus the revised Variance Theory says:

Intelligence is not the elimination of uncertainty. Intelligence is the disciplined expansion of usable structure inside irreducible uncertainty.

14. Energy-Constrained Intelligence

14.1 Physical Computation

A policy cannot be evaluated independently of its physical cost. Let

$$\mathcal{C}_{\text{comp}}(\pi)$$

be computational work, and let

$$\mathcal{J}_{\text{erase}}(\pi)$$

be irreversible information erasure. Then

$$E_T(\pi) \geq k_B T \ln 2 \cdot \mathcal{J}_{\text{erase}}(\pi).$$

Thus no intelligence can be infinitely efficient in the physical world. Computation is thermodynamic.

14.2 Energy-Time-Intelligence Trade-Off

Define an efficiency scalar only as a projection of the Pareto vector:

$$\eta_T(\pi) = \frac{C_T(\pi) + P_T(\pi) + U_T(\pi) + B_T(\pi)}{E_T(\pi)T_T(\pi)(1 + R_T(\pi))}.$$

This scalar is useful for comparison but not foundational. The foundational object is the Pareto frontier:

$$\mathcal{P}_T.$$

An apparently inefficient policy may be intelligent if it increases robustness or preserves optionality under future variance. This explains why redundancy, diversity, and exploratory behavior may persist despite short-term inefficiency.

15. Causal Emergence and Scale

Causal structure is often clearer at macroscales than microscales. Hoel's causal-emergence work argues that macro descriptions can sometimes be more informative than fully detailed micro descriptions because they better capture effective causal structure. ([MDPI](#))

Variance Theory incorporates this by defining δ relative to a scale:

$$P_{\delta, \Pi}(s' | s) = P(\Pi(x_{t+1}) = s' | \Pi(x_t) = s).$$

Different coarse-grainings Π induce different transition kernels and different causal graphs. The correct level is not always the microscopic one. The correct level is the one that maximizes predictive, compressive, and causal efficacy under energetic and observational constraints.

16. Empirical and Theoretical Predictions

16.1 Prediction 1: Complexity Peaks Between Order and Randomness

In families of systems parameterized by disorder λ , Variance Theory predicts that organized variance complexity

$$\Omega_T(\lambda)$$

will be low in ordered regimes, low or shallow in maximally random regimes, and high near edge-of-chaos regimes.

This can be tested in cellular automata, recurrent neural networks, biological regulatory networks, and adaptive agent simulations by measuring effective complexity, logical depth proxies, statistical complexity, and prediction-control performance.

16.2 Prediction 2: Intelligence Tracks Pareto Expansion, Not Raw Reward

A system that maximizes short-term reward but collapses robustness, diversity, or model-improvement capacity is less intelligent in the Variance Theory sense than a system that expands its Pareto frontier.

This predicts that highly adaptive agents should maintain exploratory and redundant substructures even when these reduce immediate scalar efficiency.

16.3 Prediction 3: Irreversible Computation Has Energetic Signature

Learning systems that compress or erase internal states should exhibit thermodynamic costs bounded below by Landauer-like constraints. This is already supported in simple physical memory systems, but Variance Theory predicts analogous constraints for increasingly complex adaptive substrates. ([Nature](#))

16.4 Prediction 4: Randomness Residuals Should Shrink Under Model Improvement but May Plateau

Given a hierarchy of increasingly powerful models $\mathcal{M}_n$, apparent residual surprise

$$R_t^{(n)}$$

should decrease with model capacity. If it plateaus at a positive asymptotic rate across physically admissible model classes, the system exhibits progressive true randomness.

17. Limitations

First, Variance Theory is a unifying formal framework, not yet a complete physical theory. It does not replace quantum field theory, general relativity, thermodynamics, or computational learning theory. It proposes a common transition-theoretic language connecting them.

Second, the progressive true-randomness rate ρ_{TR} is not generally computable, because it depends on limiting algorithmic incompressibility. Like Kolmogorov complexity itself, it is primarily a theoretical quantity requiring empirical proxies.

Third, the framework does not solve the measurement problem in quantum mechanics. It remains compatible with multiple interpretations: objective collapse, many-worlds, Bohmian mechanics, and operational decoherence. What changes across interpretations is whether P_{TR} is fundamental, branch-relative, or epistemic.

Fourth, the framework does not claim that intelligence is inevitable. It claims only that, once structures capable of compression, prediction, and utilization appear, those capacities can increase persistence under variance.

Fifth, “complexity” remains plural. Kolmogorov complexity, Shannon entropy, effective complexity, logical depth, statistical complexity, and causal emergence measure different aspects of structure. Variance Theory uses them in complementary roles rather than reducing all complexity to one scalar.

18. Conclusion

The revised Variance Theory can be summarized as follows.

1. **Variance** is physically realized state transition.
2. δ is not merely a deterministic function but a progressively randomizing transition kernel.
3. **Time** is the quasi-metric or cost structure of transition.
4. **Causality** is the directed topology of transition reachability.
5. **Energy** is the physical generator, conservation constraint, and thermodynamic cost of transition.
6. **Complexity** is not pure randomness but organized structure between rigid order and unstructured disorder.
7. **Intelligence** is the expansion of the Pareto frontier of compression, prediction, utilization, robustness, energy efficiency, time efficiency, and risk management.
8. **Evolutionary persistence** does not imply cosmic purpose; it follows from differential survival of structures better able to manage variance.
9. **Open-endedness** follows conditionally from progressive true randomness, computational irreducibility, Gödelian non-closure, and Kolmogorov incompressibility.

The deepest thesis is therefore not that the universe has a goal. It is this:

In a world of physically constrained transition, structures that discover, compress, predict, and utilize variance gain persistence. Intelligence is the continuing expansion of that capacity under irreducible uncertainty.

變易的度規湧現為時間。

變易的結構湧現為因果。

時間與因果的交互作用湧現為複雜性。

複雜性的長期演化湧現為智慧。

能量的本質是其為變易的驅動。

時間的本質是其為變易的度規。

因果的本質是其為變易的拓撲。

智慧的本質是其為變易的方向選擇權衡取捨妥協自然湧現。

真隨機性 物理層 變易不可窮盡。

計算不可約性 計算層 變易不可壓縮為零。

哥德爾不完備性 邏輯層 變易結構不可自我閉合。

Kolmogorov 不可壓縮性 信息論層 變易永遠有不可壓縮的殘餘，其極限是真隨機。

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